

Optimal energy management of a renewable microgrid integrating water supply systems

Quan Sui, Fanrong Wei, Xiangning Lin, Zhengtian Li*

State Key Laboratory of Advanced Electromagnetic Engineering and Technology, Wuhan, Hubei 430074, China

ARTICLE INFO

Keywords:

Microgrid
Water supply system
Water supply safety
Two-stage dispatching strategy

ABSTRACT

The interdependence between the power grid and water supply system (WSS) provides opportunities for energy conservation. However, the existing water supply flexibility has not been fully developed, resulting in an unsatisfactory effect. To address the problem, this paper proposes an optimal energy management strategy for a renewable microgrid (MG) integrating WSSs. First, a novel supply model that evaluates the interdependence between the large WSS and several islanded sub-systems, hydraulic stability, and desalination characteristic is established, in order to reduce the hydraulic coupling constraints and ensure water supply safety. To achieve an optimal performance of the MG integrating WSSs, a novel two-stage dispatching method is designed, where the day-head decisions fully consider the real-time power fluctuation, and the real-time stage corrects renewable generation deviation and supports external grid. Additionally, a two-stage algorithm is proposed to efficiently solve the scheduling problem. Simulation studies on a MG integrating WSSs indicate the proposed energy management strategy can reduce the water supply loss by 5.2%, and improve the MG economy by 19.5%, while supporting the external grid (EG).

1. Introduction

Water supply systems (WSSs) are responsible for seawater desalination, and living water distribution in the offshore island communities. It is indicated that a lot of electricity is consumed by WSSs in many offshore microgrids (MGs) [1,2]. Due to the presence of water tanks (WTs) or reservoirs, water supply loads (such as desalination and pumping loads) can be flexibly adjusted [3]. Therefore, if properly managed, it can not only significantly reduce the loss of water supply, but also greatly improve the operation economy of MG.

1.1. Literature review

There have been many research efforts on optimizing water supply for economic or security purposes. For example, [4,5] develop a Lagrangian relaxation-based algorithm for a collaborative water supply chain network design problem that coordinates the coordinate the water supply and wastewater collection. Ref. [6] presents a water resource and distribution system design optimization model with many objectives including minimizing operating costs while maximizing engineering performance. Ref. [7] discusses an energy efficiency evaluation measure for WSS designs and a mathematical model for optimizing water pump (WP) operation through the arrangement of WTs in a

building. Mitigation and control strategies are studied in [8] to minimize the energy consumption of the desalination device (DD) considering environmental factors. A robust optimization framework for urban bulk WSSs is proposed in [9] to explore the trade-off between efficiency and sensitivity to different climate change scenarios. A two-stage stochastic model is presented in [10,11] to reduce the energy consumption of WPs considering water demand and rainfall uncertainties. A price-dependent load management method is used in [12] to optimize the municipal water supply and wastewater treatment. Likely, an optimal water distribution model is developed in [13] to optimize the WPs according to the dynamic electricity price.

Although a lot of work has been done in the development of water supply flexibility, due to pipe hydraulic coupling constraints, the WPs and WTs still cannot be fully regulated, resulting in serious water distribution loss. Additionally, due to the lack of a feasible power-output model reference, DDs generally operate at rated power to ensure the freshwater supply, which causes a waste of power regulation capacity.

As a further study, the energy-water management in cost saving and consumption reduction has attracted researchers' interest [14]. A system-based framework is proposed in [15] to synthesize the interwoven connections between energy consumption and water use in a city. A bond graph of system modeling language is used in [16] to analyze the interdependence between the power grid and WSS. A fuzzy

* Corresponding author.

E-mail address: 453874933@qq.com (Z. Li).

<https://doi.org/10.1016/j.ijepes.2020.106445>

Received 9 April 2020; Received in revised form 5 July 2020; Accepted 8 August 2020

Available online 17 August 2020

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Nomenclature		Ω_t	vector of WP speeds
Abbreviations		P_{ij}^{in}	MG power purchased from the EG
MG	Microgrid	P_{ij}^{out}	MG power sold to the EG
WSS	water supply system	u_t^{in}	MG electricity purchase flag
DE	desalination device	u_t^{out}	MG electricity sale flag
EG	external grid	F_d	MG operational cost
WP	water pump	Parameters	
WT	water tank	A	nodal correlation matrix before system splitting
MPC	model predictive control	Γ	node-WT incidence matrix
PV	Photovoltaic	B	node-WP incidence matrix
MILP	mixed-integer linear programming	$\underline{Q}$	vector of minimum pipe water flows
Indices		$\bar{Q}$	vector of maximum pipe water flows
t, j	index for time	G	matrix of the pipe resistance parameters
DD	index for desalination device	$\underline{H}$	vector of minimum nodal pressure heads
WP	index for WP	$\bar{H}$	vector of maximum nodal pressure heads
PV	index for PV	$\Delta \underline{H}$	vector of up ramp limits of nodal pressure heads
Wind	index for wind	$\Delta \bar{H}$	vector of down ramp limits of nodal pressure heads
Re	index for renewable energy	$\underline{L}$	vector of minimum WT flows
R	index for water supply	$\bar{L}$	vector of maximum WT flows
D	index for water demand	$\underline{Q}^R$	vector of minimum water supply from the sources
In	index for the purchasing power from EG	$\bar{Q}^R$	vector of maximum water supply from the sources
Out	index for the sale power to EG	$\underline{V}$	vector of minimum water storage
Load	index for load	$\bar{V}$	vector of maximum water storage
M	index for the MPC	P^{DDN}	vector of DDs' rated power
T	index for the dispatching cycle	ΔP^{DD}	vector of down ramp limits of desalination power
exp	index for the expectation	$\Delta \bar{P}^{DD}$	vector of up ramp limits of desalination power
Λ	Day-head decision	$Q^{DD,k}$	vector of desalination output at discrete power
κ	number of breakpoint k	P^{WPN}	vector of WPs' rated power
iter	iteration number	ΔP^{WP}	vector of down ramp limits of WP power
Variables		$\Delta \bar{P}^{WP}$	vector of down ramp limits of WP power
I_t	split control matrix of water supply systems	a, b, c, d	WP parameters
Z_t	nodal correlation matrix after system splitting	$\prod(\cdot)$	unit sign function
Q_t^R	vector of water supply from the sources	P_{ij}^{PV}	PV power at hour t, minute j
Q_t^D	vector of the water demand	P_m^{PV}	maximum PV output
Q_t	vector of water flow of pipes	α_1, β_1	PV output error parameter
L_t	vector of the flow from or to WTs	P_{ij}^{Wind}	wind power at hour t, minute j
V_t	vector of the water storage	P_m^{Wind}	maximum wind generation
V_t^1	vector of the water storage of WTs	α_2, β_2	wind generation error parameter
V_t^2	vector of the water storage of reservoirs	p_t^{in}	electricity purchase price
H_t	vector of nodal hydraulic pressure heads	p_t^{out}	electricity sale price
P_t^{DD}	vector of desalination power	$\bar{P}^{GD}$	maximum exchange power between the EG and MG
Q_t^{DD}	vector of desalination outputs	P_{ij}^{Load}	electric load
γ_t^k	vector of desalination model linearization	V_{exp}	expected water storage at the end
β_t^k	vector of desalination model linearization	ΔP_{ij}^{Re}	forecasting error of renewable energy generation
P_t^{WP}	vector of WP power	M, E, Y	coefficient matrixes
		X	the matrix of variables x_t
		iter_m	maximum iterative number
		ε	allowed error

decision based energy dispatch for offshore industrial MGs is designed in [17] to coordinate desalination process and multi-stage DGs. A security constrained unit commitment problem that optimizes the commitment and schedule of power generators and DDs is formulated in [18]. A novel scheduling strategy that coordinates the DD and batteries is presented in [19]. Ref. [20] proposes a network-constrained unit commitment model that integrates and optimizes the energy flexibility of water distribution systems in the day-ahead operation of power systems. Ref. [21] presents a real-time rolling horizon energy management method for stand-alone wind-powered-desalination MGs to maximize wind energy utilization and minimize diesel generation. Ref. [22] discusses an optimal two-step demand response scheduling

framework where the first step optimizes the WPs and WTs for minimizing the water procurement cost, and the second step optimizes the demand response for maximizing the service profit. Ref. [23] designs a water-electricity combined supply method for building-integrated-photovoltaics communities to optimize the operational costs and power fluctuation on multiple time scales. An optimal pump demand response scheduling problem is formulated as a mixed-integer optimization problem and solved using a branch and bound algorithm in [24]. A new mathematical programming approach is proposed [25] for WP scheduling in a common class of branched networks with one pumping station raising water to elevated tanks. We can learn much from the above literature.

However, the frequent use of WPs to smooth electric power will cause water pressure fluctuations on the minute-time scale, resulting in a bad user experience, while without a feasible model reference, direct regulation of desalination load will cause desalination loss. Even worse, the classic day-head dispatching model cannot consider the real-time electric and hydraulic power fluctuation on the minute-time scale. Therefore, following the day-head decisions, both the DDs and WPs may not smooth the instantaneous renewable generation variation due to power ramp constraints, and the real-time water pressure fluctuation may exceed the allowance. Additionally, regarding the MG connected to the external grid (EG), the WSSs can not only respond to the local MG, but also support EG (reduce the demand for peak and frequency modulation), which has not been discussed.

1.2. Contribution

It can be seen from the literature review that the water supply loads are not well studied on flexibility development and energy saving. Therefore, a novel water-energy management strategy needs to be designed.

Note that there are two main obstacles to solve the above problem. **Firstly**, a novel water supply model needs to be proposed, in order to release the regulation ability. **Secondly**, a novel dispatching method needs to be presented to ensure the performance of the MG integrating WSSs.

To tackle the first challenge, we propose that by controlling pipe valves, the large WSS can be split into several islanded sub-systems to reduce nodal coupling constraints. The nodal hydraulic relationship before and after the system splitting is modelled. Also, the hydraulic stability and desalination characteristic are analyzed to ensure water supply safety. To tackle the second challenge, we propose a novel two-stage dispatching method. The day-head stage fully considers the hydraulic fluctuation, and instantaneous load regulation difficulty due to the real-time power fluctuation, and the real-time stage corrects renewable generation deviation and supports EG. Hence, the main contributions of this paper are summarized below:

- (1) A novel water supply model is proposed, where the interdependence between the large WSS and several islanded sub-systems, and the hydraulic stability and desalination characteristic are analyzed.
- (2) A novel two-stage MG dispatching strategy is designed, where the day-head decisions fully consider the electric and hydraulic fluctuation using Latin hypercube sampling method, while the real-time stage corrects renewable generation deviation and supports EG.
- (3) A feasible two-stage algorithm is designed to solve the optimization problem efficiently.
- (4) Using the realistic demand and renewable generation, the effectiveness and benefits of the proposed energy management strategy are illustrated.

The rest of this paper is organized as follows: Materials and Methods are presented in Section 2. Experimental results and discussion are introduced in Section 3. The conclusion is drawn in Section 4.

2. Materials and methods

Fig. 1 shows the typical structure of a renewable MG integrating WSSs. Due to the existence of tanks and reservoirs, the water supply load can respond to the MG and EG demand while satisfying the hydraulic usage demand. To achieve an optimal performance of the MG integrating WSSs, a novel water-energy management strategy is proposed in this paper.

2.1. Novel water supply model

Fig. 2 shows a novel water supply mode. The WSS consists of a DD, reservoirs, WPs, WTs, water pipes, and water usage loads. Living water is produced by the DD and stored in the reservoir. WPs can distribute water in the source to the hydraulic load, and WTs are used to regulate the pipe flow by storing or releasing water. According to the flow law, the hydraulic factors (such as pressure heads) of adjacent nodes are connected to each other. To reduce the hydraulic coupling constraints, the large WSS can be split into several islanded sub-systems by controlling pipe valves. The detailed model is presented as follows:

1) Large system split model

The split of the WSS will impact the nodal correlation matrix and water flow constraints, as follows:

$$Z_t = I_t A \quad (1)$$

$$Q_t^R - Q_t^D = Z_t Q_t + G^T L_t \quad (2)$$

$$B Q_t \geq 0 \quad (3)$$

$$\underline{Q} \leq Q_t \leq \bar{Q} \quad (4)$$

where (1) expresses the nodal correlation relationship before and after the large system splitting, (2) denotes the pipe flow balance constraint, (3) represents the unidirectional flow limit for the pipeline where the WP is located, and (4) indicates the allowed water flow range. Note that the nodal correlation matrixes of the WSS in Fig. 2 can be presented as follows:

$$\begin{bmatrix} 1 & -1 & \widehat{Z}_t & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & I_{t3} - I_{t3} & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & I_{t3} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & -1 & \widehat{A} & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

2) Hydraulic stability model

Excessive demand response of the WSS may damage the water usage experience. Therefore, the hydraulic stability factors such as pressure fluctuation must be considered while carrying out the demand response.

$$V_t = \Delta \Gamma H_t \quad (5)$$

$$-Z_t H_t = G \prod (Q_t) |Q_t|^{1.852} \quad (6)$$

$$\underline{H} \leq H_t \leq \bar{H} \quad (7)$$

$$\Delta \underline{H} \leq H_t - H_{t-1} \leq \Delta \bar{H} \quad (8)$$

where (5) denotes the water pressure of the tanks and reservoirs, (6) expresses the pressure head loss due to pipe frictions [26,27], (7)–(8) represent the allowed pressure head value and pressure fluctuation range.

3) Energy storage model

The WT and reservoir can regulate the pipe flow by storing or

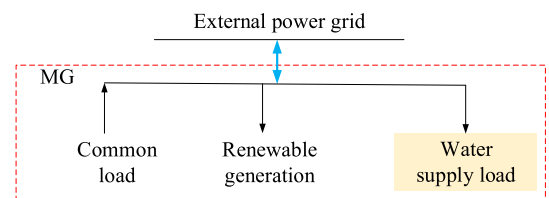


Fig. 1. Structure of a renewable MG integrating WSSs.

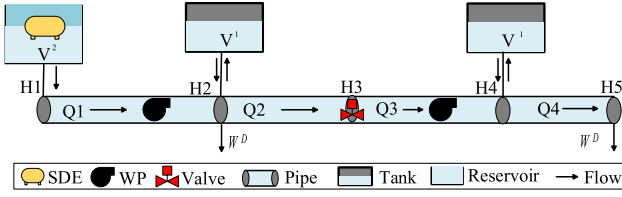


Fig. 2. Novel water supply mode.

releasing water, as follows:

$$\underline{L} \leq L_t \leq \bar{L} \quad (9)$$

$$V_t^1 = V_{t-1}^1 + L_t \quad (10)$$

$$\underline{Q}^R \leq Q_t^R \leq \bar{Q}^R \quad (11)$$

$$V_t^2 = V_{t-1}^2 + Q_t^{DD} - Q_t^R \quad (12)$$

$$V_t = [V_t^1, V_t^2] \quad (13)$$

$$\underline{V} \leq V_t \leq \bar{V} \quad (14)$$

where the input/output flow and water storage variation constraints of WTS are expressed in (9)–(10), the output flow and water storage variation limits of water sources are represented in (11)–(12), the water storage volumes are constrained to (13)–(14).

4) Controllable load model

Both the desalination and pumping loads of the WSS are controllable. Their operational models are needed for energy management. The power-output characteristic is modelled by analyzing the internal desalination mechanism. To increase the readability, the detailed analyses are depicted in Appendix, and Fig. 3 directly shows the result. Linearized techniques are used to mathematically depict the power-output characteristic, as shown in (15)–(22). Note that the obtained desalination characteristic is consistent with the experimental results on discrete conditions [28], implying the feasibility and correctness of the derived model.

$$0 \leq P_t^{DD} \leq P^{DDN} \quad (15)$$

$$\Delta P_t^{DD} \leq P_t^{DD} - P_{t-1}^{DD} \leq \Delta \bar{P}^{DD} \quad (16)$$

$$P_t^{DD} = \sum_{k=1}^K \frac{k}{K} P^{DDN} \gamma_t^k \quad (17)$$

$$Q_t^{DD} = \sum_{k=1}^K Q^{DD,k} \gamma_t^k \quad (18)$$

$$\sum_{k=1}^{K-1} \beta_t^k = 1 \quad (19)$$

$$\sum_{k=1}^K \gamma_t^k = 1 \quad (20)$$

$$\gamma_t^k \leq \beta_t^{k-1} + \beta_t^k \quad (21)$$

$$0 \leq \gamma_t^k \leq 1 \quad (22)$$

where (15)–(16) denote the allowed desalination power and ramp rate constraints, (17)–(18) express that the actual desalination power and output are linear combinations of two adjacent fixed power and output points with weights γ_t^k and γ_t^{k+1} , (19)–(22) limit the active adjacent power points with binary variables β_t^k .

Additionally, the WP model is presented as follows:

$$0 \leq P_t^{WP} \leq P^{WPN} \quad (23)$$

$$\Delta P_t^{WP} \leq P_t^{WP} - P_{t-1}^{WP} \leq \Delta \bar{P}^{WP} \quad (24)$$

$$P_t^{WP} = \Omega_t^3 \odot (d1^U - e((1^U \oslash \Omega_t) \odot BQ_t)) \quad (25)$$

$$BZ_t^T H_t = \Omega_t^2 \odot (a1^U - b((1^U \oslash \Omega_t) \odot BQ_t)^c) \quad (26)$$

where the WP power and ramp rate constraints are denoted in (23)–(24), the relationships between the WP power, speed and driven flow are represented in (25)–(26) [26,27]. Note that the nonlinear formulas can be linearized according to [20,22].

To summarize, the novel water supply model consisting of linear constraints can increase the regulation flexibility and operational stability. It can improve the demand response performance according to the following dispatching strategy.

2.2. Two-stage dispatching strategy

To ensure the performance of the MG integrating WSSs, a novel two-stage dispatching strategy is proposed, as observed in Fig. 4. Fully considering the real-time electric and hydraulic fluctuation, the day-head stage optimizes the WSS to minimize the MG operational cost, and the real-time stage mitigates the MG renewable power fluctuation and respond to the EG demand. We can see the WSS can not only satisfy the MG demand, but also support the EG.

(1) Day-ahead dispatching stage

Based on an hour scheduling step, the classic day-head dispatching model cannot consider the real-time power fluctuation. However, the violent variation of renewable generation on the minute-time scale may exceed the ramp ability of the controllable loads in (16) and (24). Also, the hydraulic stability constraint of (8) may not be satisfied. To solve the problem, a novel day-head model considering the real-time power fluctuation is proposed, as follows:

First, the real-time power fluctuation is simulated in the day-head stage using Latin hypercube sampling method. As we all know, the hourly wind power P_t^{Wind} and photovoltaic (PV) output P_t^{PV} can be predicted in the day-head stage, and the power fluctuation law has been studied, as shown in (27)–(28) [19,29]. Therefore, we can use Latin hypercube sampling method to simulate the real-time power on the minute-time scale. For example, the real-time power on a 2-min time scale can be obtained if 30 points are sampled per hour. Define P_j^{PV} and P_j^{Wind} denote the PV output and wind power on hour t , minute j . Note that compared with Monte Carlo sampling [19], Latin hypercube sampling can obtain more comprehensive power distribution, which can better ensure load ramping and hydraulic stability.

$$f(P_t^{PV}) = \frac{1}{P_m^{PV} B(\alpha_1, \beta_1)} \left(\frac{P_t^{PV}}{P_m^{PV}} \right)^{\alpha_1-1} \left(1 - \frac{P_t^{PV}}{P_m^{PV}} \right)^{\beta_1-1} \quad (27)$$

$$f(P_t^{Wind}) = \frac{1}{P_m^{Wind} B(\alpha_2, \beta_2)} \left(\frac{P_t^{Wind}}{P_m^{Wind}} \right)^{\alpha_2-1} \left(1 - \frac{P_t^{Wind}}{P_m^{Wind}} \right)^{\beta_2-1} \quad (28)$$

The day-head dispatching problem can be presented as follows:

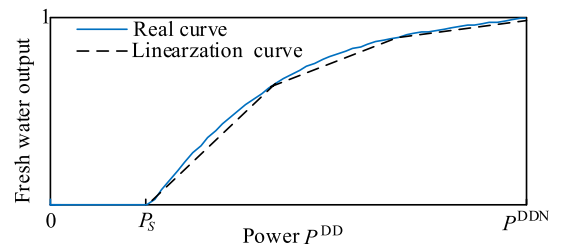


Fig. 3. Desalination characteristic.

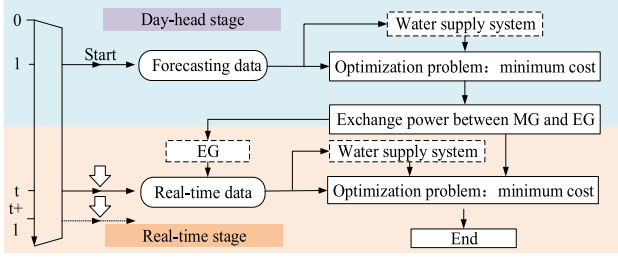


Fig. 4. Diagram of two-stage MG dispatching strategy.

$$F_d: \min \sum_{t=1}^T \sum_{j=1}^J (P_{ij}^{in} \cdot P_t^{in} - P_{ij}^{out} \cdot P_t^{out}) \quad (29)$$

$$\text{s.t. } 0 \leq P_{ij}^{in} \leq u_t^{in} \bar{P}^{GD} \quad (30)$$

$$0 \leq P_{ij}^{out} \leq u_t^{out} \bar{P}^{GD} \quad (31)$$

$$u_t^{in} + u_t^{out} \leq 1 \quad (32)$$

$$P_{ij}^{in} - P_{ij}^{out} = P_{ti}^{in} - P_{ti}^{out} \quad \forall i, j \quad (33)$$

$$P_{ij}^{PV} + P_{ij}^{Wind} + P_{ij}^{in} - P_{ij}^{out} = P_{ij}^{WP} + P_{ij}^{DD} + P_{ij}^{Load} \quad (34)$$

$$V_0 = V_{exp} \quad (35)$$

where the objective is denoted in (31), the exchange power is constrained to (30)–(33), the power balance is limited in (34), the energy storage is constrained to (35).

(2) Real-time dispatching stage

In the real-time stage, the EG may have power demand for peak shaving or frequency modulation. This demand can be reflected in the real-time electricity price. By correcting the exchange power, the MG integrating a WSS can provide power support for the EG. Based on the above consideration, the model predictive control (MPC) method is used to coordinate the MG energy arbitrage and EG power support simultaneously. The detailed model is presented as follows:

$$F_{rd}: \min \sum_{\tau=t}^{t+M-1} \sum_{j=1}^J (P_{\tau}^{in} \cdot P_{\tau}^{in} - P_{\tau}^{out} \cdot P_{\tau}^{out}) \quad (36)$$

$$\text{s.t. } V_{t+M-1} = \hat{V}_{t+M-1} \quad (37)$$

$$\Delta P_{ij}^{Re} = P_{ij}^{PV} + P_{ij}^{Wind} - \hat{P}_{ij}^{WP} - \hat{P}_{ij}^{Wind} \quad (38)$$

$$\begin{cases} P_{ij}^{WP} + P_{ij}^{DD} > \hat{P}_{ij}^{WP} + \hat{P}_{ij}^{DD} + \Delta P_{ij}^{Re}, & \text{if } \Delta P_{ij}^{Re} < 0 \\ P_{ij}^{WP} + P_{ij}^{DD} < \hat{P}_{ij}^{WP} + \hat{P}_{ij}^{DD} + \Delta P_{ij}^{Re}, & \text{if } \Delta P_{ij}^{Re} > 0 \end{cases} \quad (39)$$

where $\hat{\cdot}$ marks the day-ahead prediction, the real-time dispatching objective is denoted in (36), the SOC that follows the day-ahead decisions is constrained to (37), the controllable load constraints are expressed in (38)–(39). Three scenarios are analyzed as follows:

- 1) Scenario 1 ($\Delta P_{ij}^{Re} < 0$). When the real-time purchased-electricity price is reduced (the MG is encouraged to purchase more power from the EG), the MG can increase the exchange power by increasing controllable load power according to (38). If the purchased-electricity price is increased (the MG is encouraged to purchase less power from the EG), the MG can maintain the day-head exchange power by reducing controllable load power, rather than increase the EG burden.
- 2) Scenario 2 ($\Delta P_{ij}^{Re} > 0$). For a higher real-time electricity price, the MG can reduce the exchange power by decreasing controllable load power according to (38). On the other hand, for a lower electricity

price, the MG can maintain the exchange power by consuming more renewable generation.

- 3) Scenario 3 ($\Delta P_{ij}^{Re} = 0$). The MG can directly support the EG according to the real-time electricity price.

2.3. Solution algorithm

The above dispatching model consists of linear objectives and constraints. Therefore, it can be converted into a classic mixed-integer linear programming (MILP) problem. The day-head dispatching problem can be presented as follows:

$$F_d: \min MX$$

$$\text{s.t. } EX = Y$$

$$X = [X1, X2]$$

$$X1 = [x_1, x_2, \dots, x_T]$$

$$X2 = [x_1, x_2, \dots, x_T] \quad (40)$$

where M , E and Y express the coefficient matrixes, X denotes the variables matrix, where $X1$ represents the matrix of the fast-action variables x_t . ($x_t = [x_{t1}, x_{t2}, \dots, x_{tJ}]$), $X2$ expresses the matrix of the slow-action variables x_t .

However, a large number of binary variables make it hard to efficiently solve this problem. To speed up the solution, a two-stage method is proposed in this paper, as follows:

In the first stage, the slow-action variables are determined. With $J = 1$ of (40), the classic day-head dispatching model can be obtained. By optimizing this model, we can obtain the slow-action values such as split control variables I_t , and exchange power control variables u_t^{in}/u_t^{out} . These values are called $X2^*$ and passed to the second stage.

In the second stage, with $J = 30$ of (40), the proposed day-head dispatching model is optimized. First, considering that the storage water are the only variables related to time series, the original dispatching problem is separated into two sub-problems with the associated variables V_t .

$$F_d: \min MX$$

$$\text{s.t. } EX = Y$$

$$X = [X1, X2^*]$$

$$X = [X1, X2^*] \quad (41)$$

$$F_d^1: \min M^1 X^1$$

$$\text{s.t. } E^1 X^1 = Y^1$$

$$V_{T/2} = V^*$$

$$X^1 = [x_1, x_2, \dots, x_{T/2}] \quad (42)$$

$$F_d^2: \min M^2 X^2$$

$$\text{s.t. } E^2 X^2 = Y^2$$

$$V_{T/2} = V^*$$

$$X^2 = [x_{T/2+1}, x_{T/2+2}, \dots, x_T] \quad (43)$$

We can see that the end water storage of the first problem is the initial storage of the second problem. Because a larger V^* will result in the increase of F_d^1 , and decrease of F_d^2 , we speculate that $F_d = F_d^1 + F_d^2$ will first reduce and then increase as V^* increases.

Based on the above analysis, we use iterative methods to solve the sub-problems. The iterative factor is set as follows:

$$V_{*}^{iter+1} = V_{*}^{iter} - \rho (F_d^{iter} - F_d^{iter-1}) / (V_{*}^{iter} - V_{*}^{iter-1}) \quad (44)$$

where ρ is the scaling factor.

The solution steps are represented as follows:

Step 1: Initialization

The renewable generation, electric and hydraulic load, WSS data and electricity price are provided. Additionally, the initial iterative number $iter = 0$, maximum iterative number $iter_m$, initial V_{*}^{iter} , and maximum error ε are set.

Step 2: The first stage solution

With $J = 1$, (40) is solved. The values of the slow-action variables are called $X2^*$ and passed to the second stage.

Step 3: The second stage solution
 With $J = 30$, (42) and (43) are solved.
 Step 4: Iterative stopping condition checking

Case 1: $\text{iter} > \text{iter}_m$
 Case 2: $|V_*^{\text{iter}+1} - V_*^{\text{iter}}| \leq \varepsilon$

The appearance of any case above will stop the iteration. Or else, go to Step 4.
 Step 4: Update V_*^{iter} .
 Calculate V_*^{iter} according to (44), and go to Step 3.

3. Experimental results and discussion

3.1. Test system

A grid-connected MG integrating a WSS is used to verify the effectiveness of the proposed strategy. The renewable generation, electric loads, hydraulic loads and electricity price are shown in Fig. 5 [30]. Fig. 6 shows the typical WSS [10]. Without loss of generality, the large WSS can be split three sub-systems marked as ①②③. The rated desalination power is set to 600 kW, and the water supply parameters are shown in Table 1 in appendix [23]. Considering the natural precipitation, the daily freshwater production is set to 2100 m³ in summer and 3150 m³ in winter, respectively.

To show the superiority of the proposed strategy, three cases are compared, as follows:

Case 1: The novel WSS model is used to carry out demand side response using the proposed dispatching strategy.
 Case 2: The WSS is dispatched to respond to the MG demand according to [23], where the WSS cannot be split and the DD always works at rated power.
 Case 3: The WSS carries out demand side response according to electricity price [22], where the WSS cannot be split and the DD works at rated power.

The simulations are coded in MATLAB and solved by Gurobi on a Windows-based PC with 8 GB of RAM.

3.2. Analysis of the energy management method

According to the above simulation parameters, the MG integrating WSSs is tested. The results are as follows:

3.2.1. Energy consumption analysis

1) Typical scenario in summer

Fig. 7 shows that the electricity consumption of the controllable loads in three cases in summer. As observed from the figure, the controllable loads in case 3 are shifted to carry out energy arbitrage by consuming more low-price electricity and less high-price electricity. Unlike this, the controllable loads in case 1 and 2 closely follow the abundant renewable generation. However, compared with those in case 2, the energy consumption of the controllable loads in case 1 is more elastic. Therefore, the controllable loads in case 1 can better carry out peak shaving and valley filling.

Fig. 8 shows the desalination power and production efficiency in three cases. For case 2 and case 3, the DDs are only allowed to work at rated power. The desalination time is optimized to reduce the system

operational cost. However, in case 1, both the working time and power of the DD are dispatched according to the desalination characteristic. To satisfy the freshwater demand of 2100 m³, 5551-kWh, 6000-kWh, and 6000-kWh electricity are consumed by DDs in three cases, that is, the freshwater productions per kilowatt hour are 0.3783 m³ in case 1, and 0.3500 m³ in case 2 and case 3, respectively. Compared with that in case 2 and 3, desalination energy consumption in case 1 can be reduced by 7.5%.

Fig. 9 depicts the WP power in three cases. ①②③ express the sub-systems, and the connection between them indicates the large system split state. As observed from the figure, the WSS in case 1 can be dynamically split by controlling valves, and the WP can be operated in the range of 38.10–398 kW. However, limited by the nodal coupling constraints, the WPs can only work within 86.40–422.2 kW in case 2 and 71.01–345.14 kW in case 3. In the dispatching period, the WPs in case 1–3 consume 3713.1-kWh, 3918.2-kWh, and 4022.0 kWh electricity, respectively. Compared with that in case 2 and 3, the water distribution loss in case 1 can be reduced by 5.2%, and 7.7%, respectively.

Fig. 10 shows the exchange power between the MG and EG in three cases. We can see that compared with that in case 2 and 3, the exchange power in case 1 is smoother. Additionally, unlike that the exchange energy is 3704.0 kWh in case 2, and 8959.3 kWh in case 3, only 2946.1-kWh electricity is traded in case 1. The results show that the MG integrating the novel WSS is more elastic and friendly to the EG.

2) Typical scenario in winter

Fig. 11 shows the energy consumption of the controllable loads in three cases in winter. The controllable load in case 3 still only responds to the low-price electricity. Although both of them can follow the renewable generation variation, the controllable load in case 1 is more elastic than that in case 2.

Fig. 12 shows the desalination performance in three cases. Only the

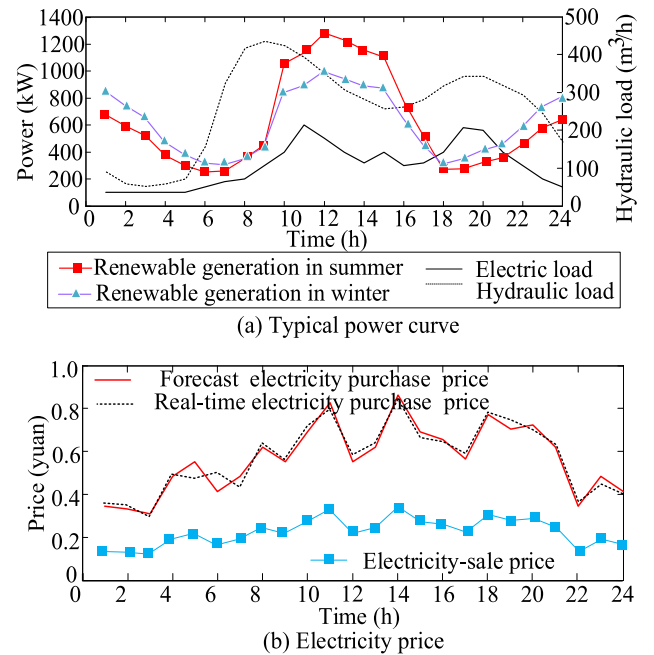


Fig. 5. Curve of the typical power and electricity price.

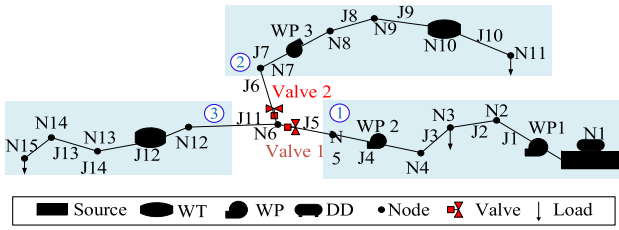


Fig. 6. Distributed water supply system.

Table 1
MG operational costs.

Season	Case	Electricity purchased		Electricity sold		Cost (Yuan)
		Electricity (kWh)	Cost (yuan)	Electricity (kWh)	Cost (yuan)	
Summer	Case 1	3007.4	1442.0	0	0	1442.0
	Case 2	3704.0	1845.4	0	0	1845.4
	Case 3	6308.9	3041.0	2650.4	483.6	2557.4
Winter	Case 1	4623.5	2308.2	0	0	2308.2
	Case 2	5460.9	2813.8	0	0	2813.8
	Case 3	7027.4	3599.4	1558.3	294.6	3304.8

desalination time can be optimized in case 2 and case 3, while both the time and power are dispatched in case 1. To produce the 3150-m³ freshwater, 8357.7-kWh, 9000-kWh, and 9000-kWh electricity are consumed in three cases, that is, the freshwater productions per

kilowatt hour are 0.3769 m³ in case 1, and 0.3500 m³ in case 2 and case 3. Therefore, the desalination loss can be reduced by 7.1% in case 1.

Fig. 13 depicts the WP power in three cases. Benefiting from the WSS split, the WPs in case 1 can be operated in the range of 42.43–404.3 kW, and the energy consumption is 3779.6 kWh. However, the WPs in case 2 can only work within 78.25–366.59 kW, and the energy loss is 4019.5 kWh. Also, the WP operational range in case 3 is 78.52–352.20 kW, and the electricity consumption is 3989.1 kWh. Compared with that in case 2 and 3, the water distribution loss in case 1 can be decreased by 6.0% and 5.3%, respectively.

Fig. 14 depicts the exchange power between the MG and EG. Compared with the exchange electricity of 5460.9 kWh in case 2 and 8585.7 kWh in case 3, the exchange electricity in case 1 is reduced to 4609.1 kWh. Also, the exchange power in case 1 is smoother. The results indicate that the novel WSS can make the exchange power more elastic and friendly to the EG.

3.2.2. Power smoothing analysis

Taking the typical scenario in summer as an example, the real-time MG performances are analyzed, as follows:

Fig. 15 depicts the real-time electric and hydraulic balance in case 1. In this case, the power fluctuations are smoothed by the DD and WPs, while most of hydraulic pressure heads remain stable.

Fig. 16 shows the real-time power balance in case 2. In this case, the regulation ability of the control loads is not fully developed. Although the WSS is used to carry out demand side response, both the exchange power and pressure head fluctuate.

Fig. 17 shows the power balance in case 3. The hydraulic stability is

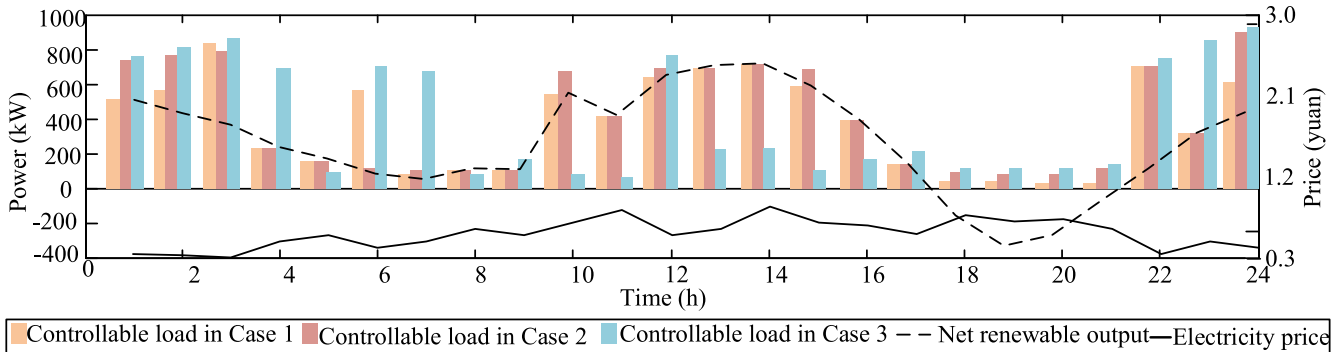


Fig. 7. Electricity consumption of the controllable loads in three cases in summer.

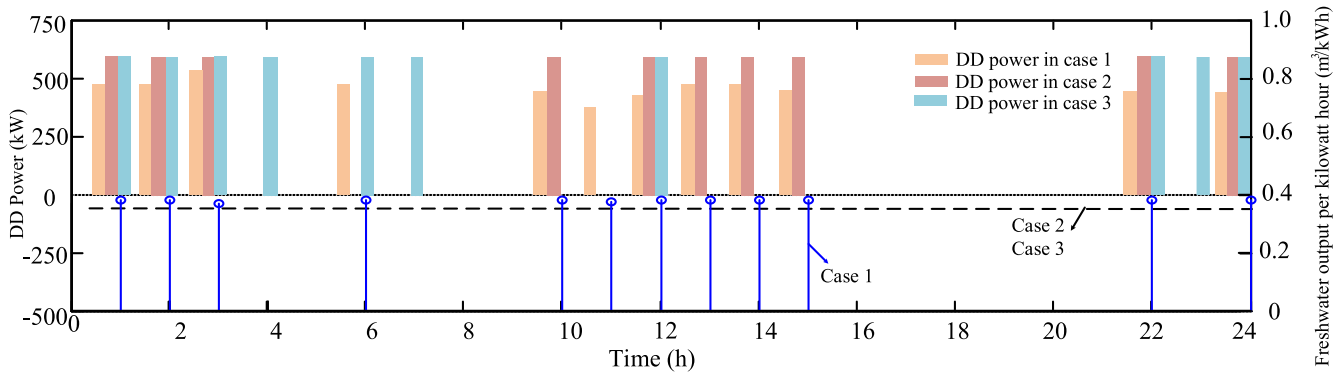


Fig. 8. Desalination power and production efficiency in three case 3 in summer.

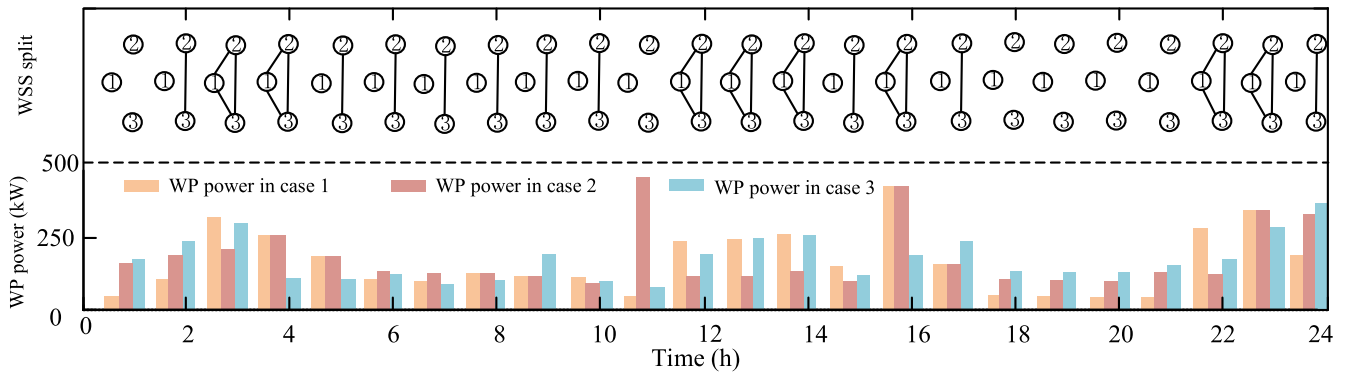


Fig. 9. WSS split and WP power in three case 3 in summer.

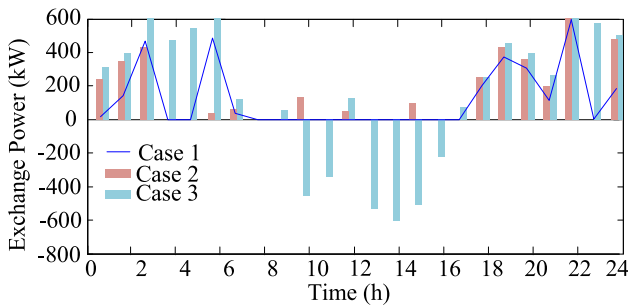


Fig. 10. Exchange power between the MG and EG in summer.

ensured. However, the exchange power fluctuates all the time.

To summarize, the WSS in case 1 can well conduct on the demand side response while ensuring the hydraulic stability.

3.3. Economy analysis

Table 1 shows the MG operational economy in three cases. As observed from the table, the typical daily operational costs in summer are 1442.0 yuan in case 1, 1845.4 yuan in case 2 and 2557.4 yuan in case 3. Likely, in a winter typical day, the MGs for three cases spend 2308.2 yuan, 2813.8 yuan and 3304.8 yuan in supplying electricity, respectively.

Suppose that there are 180 days in summer and winter every year.

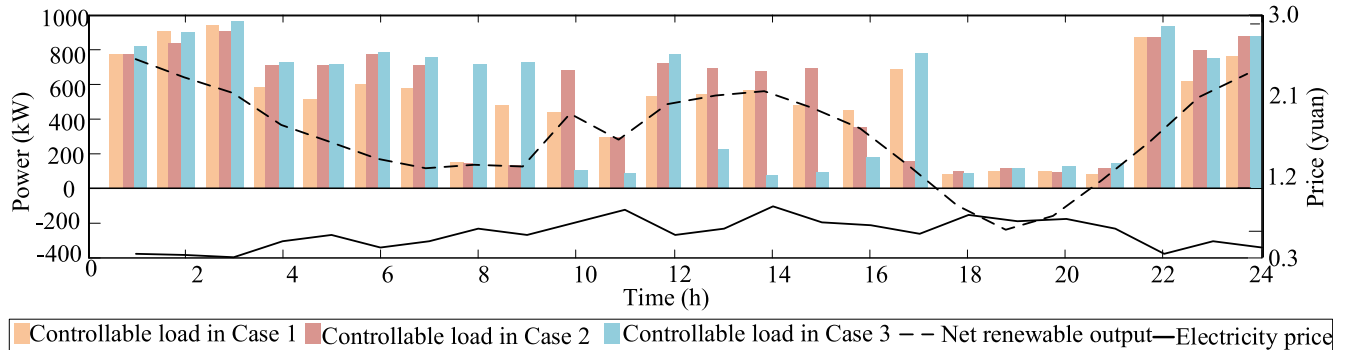


Fig. 11. Electricity consumption of the controllable loads in winter.

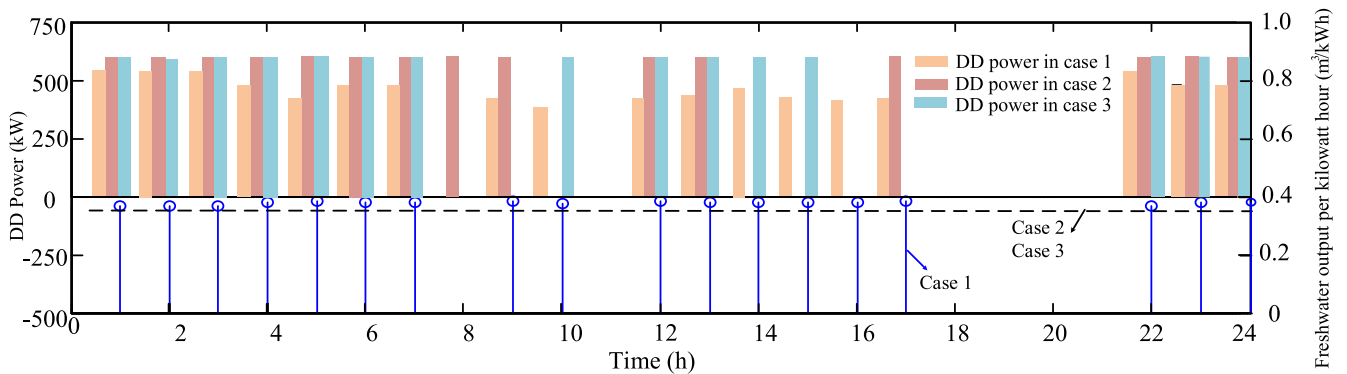


Fig. 12. Desalination power and production efficiency in three case 3 in winter.

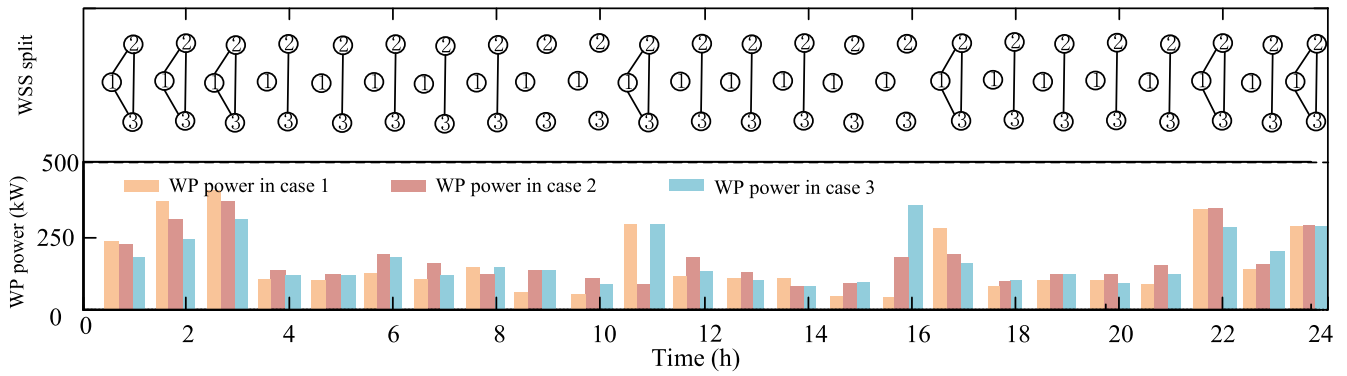


Fig. 13. WSS split and WP power in three case 3 in winter.

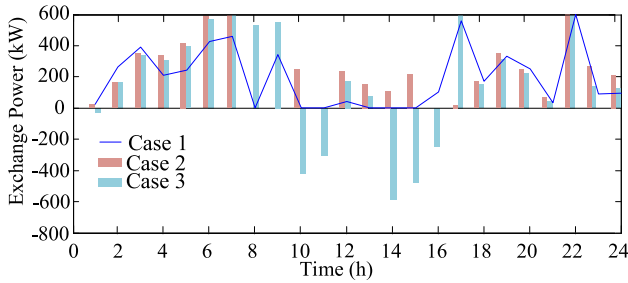


Fig. 14. Exchange power between the MG and EG in winter.

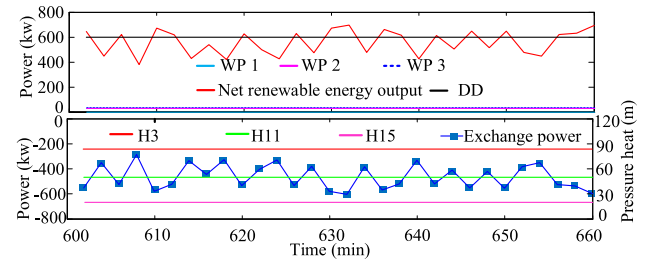


Fig. 17. Real-time electric and hydraulic balance in case 3.

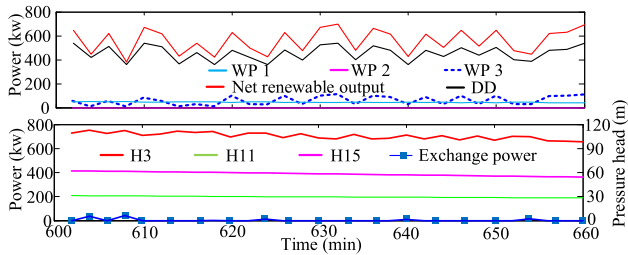


Fig. 15. Real-time electric and hydraulic balance in case 1.

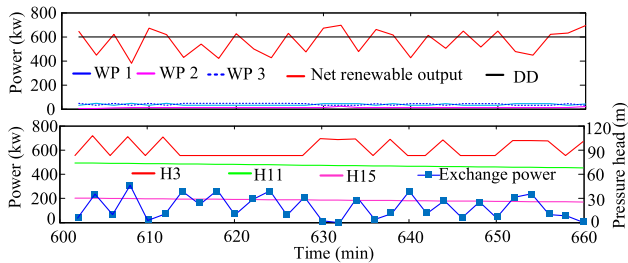


Fig. 16. Real-time electric and hydraulic balance in case 2.

The annual MG costs are 675,036 yuan in case 1, 838,660 yuan in case 2, 1,055,200 yuan in case 3, respectively. Compared with that in case 2 and case 3, the MG cost in case 1 can be reduced by 163,624 yuan (19.5%), and 380,164 yuan (36.0%), respectively. The results indicate that the proposed strategy in this paper has the best economy.

4. Conclusion

This paper proposes an optimal energy management strategy for a

renewable MG integrating WSSs. A novel water supply model is presented that considers the large system split feasibility, hydraulic stability, and desalination characteristic. To ensure the performance of the MG integrating WSSs, a two-stage scheduling method is designed for minimizing the MG operational costs and power fluctuations. This scheduling problem is efficiently solved by a two-stage algorithm. The simulation study indicates that the energy-water management strategy can carry out energy arbitrage, power smoothing while supporting the external grid simultaneously.

The novel water supply model can significantly improve the operational flexibility of the DD and WP. The desalination loss and water distribution loss can be reduced by more than 7.1% and 5.2%, respectively. Additionally, using the proposed scheduling strategy, the operational cost of the MG integrating WSSs can be reduced by 19.5%. The hydraulic and electric stability can also be ensured.

However, some technical details neglected in the theoretical discussion of this paper may be of great importance in practical application. Therefore, we will conduct a real case study in the future. Additionally, the extreme weather significantly affects the reliability of the power supply. The resilience of power grid integrating water supply systems will also be our future work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the State Grid Corporation of China (5700-201999495A-0-0-00).

Appendix 1

See Table 2.

Table 2
Water supply pipe parameters.

Pipe	Length (m)	Diameter (m)
J1	18	0.15
J2	401	0.23
J3	18	0.30
J4	1087	0.40
J5	1002	0.24
J6	846	0.24
J7	44	0.25
J8	18	0.25
J9	23	0.25
J10	16	0.11
J11	1003	0.32
J12	30	0.15
J13	6	0.23
J14	13	0.30

Appendix 2

See Table 3 and Fig. 18.

To make full use of the regulation ability, the desalination characteristic is analyzed quantitatively. The DD consists of the high-pressure pump, booster pump, reverse osmosis membrane and closed recovery unit [31], as observed in Fig. 1. Q_f and Q^{DD} denote the input seawater and freshwater output, respectively. P_f and P_p express the corresponding flow pressure, respectively. Q_1 and Q_2 represent the inflow of the high-pressure pump and booster seawater, respectively. P_s and P_0 denote the pressurized seawater pressure and Atmospheric pressure, respectively. Q_c and P_c represent the flow and pressure in the recovery unit, respectively.

According to the system design equation from Dow, desalination mechanism is presented as follows [32,33]:

$$Q_p = N_E S_E A(\bar{\pi}) TCF \cdot FF \cdot \{P_f - 0.5\Delta P_{fc} - P_p - \pi_f [\bar{C}_{fc}/C_f \times C\bar{P}F - (1 - \bar{S}R)]\} \quad (1)$$

where N_E and S_E express the number and area of the membranes, $A(\bar{\pi})$ and $\bar{\pi}$ denote the hydraulic permeability coefficient and average osmotic pressure between the raw water and concentrated brine, TCF and FF represent the temperature correction coefficient and membrane fouling factor, C_f is the concentration of the raw water, $\bar{C}_{fc}$ and ΔP_{fc} are the average pressure difference and concentration between the raw water and concentrated salt water, π_f is the osmotic pressure of pretreated raw water, $C\bar{P}F$ and $\bar{S}R$ express the average concentration polarization coefficient and desalination rate of the system.

Define $\bar{q}_{fc}$ denotes the average flow of inlet and outlet water of each RO membrane pressure vessel. We can obtain the following formulas:

$$\begin{aligned} \bar{q}_{fc} &= 0.5(Q_{pV} + Q_{bPV}) = 0.5(Q_{pV} + Q_{pV} - Q_{pPV}) \\ &= Q_{pPV}/Y - 0.5Q_{pPV} \end{aligned} \quad (2)$$

$$Y = Q_p/Q_f \quad (3)$$

where Q_{pPV} represents the water production flow of each pressure vessel, Y denotes the system recovery rate.

The average pressure drop between raw water and concentrated brine meets the following relationship [34]:

$$\Delta P_{fc} = 0.01n\bar{q}_{fc}^{1.7} \quad (4)$$

where n expresses the number of membrane elements of each pressure vessel.

Table 3
Membrane parameters.

N_E (each RO membrane)	392
S_E (ft^2)	400
f (gfd)	9
π_f (psi)	328.2
$A(\pi)$ (gfd/psi)	0.125
TCF	0.8597
FF	0.60
n (each RO membrane)	7
PV (each RO membrane)	56
Q_{pPV} (gpm)	32.35
SR	99.95%

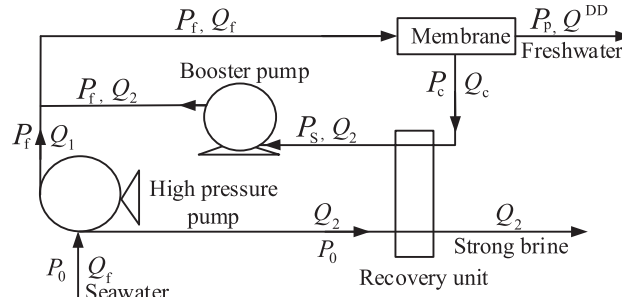


Fig. 18. Desalination mechanism.

The relationship between the concentration changes before and after the solution passes through the membrane is shown as follows:

$$\bar{C}_{fc}/C_f \approx -\ln(1 - Y)/Y \quad (5)$$

$$C\bar{P}F = \exp\{0.7[1 - (1 - Y)^{1/n}]\} \quad (6)$$

$$\bar{\pi} = \pi_f \times \bar{C}_{fc}/C_f \times C\bar{P}F \quad (7)$$

According to (1)–(7) and membrane parameters in Table 2, we can obtain a novel equation about Q_p . (8) can be further obtained by converting unit gpd to m^3/h , as follows:

$$Q^{DD} = 1.594(P_f - 0.035(32.35/Y - 16.175)^{1.7} + 328.2(\ln(1 - Y)/Y \times e^{0.7(1 - (1 - Y)^{1/7})}) - 0.0005) \quad (8)$$

On the other hand, the energy and material for the freshwater production satisfy the following balance relationship [32,33]:

$$\bar{P} = (P_f Q_1 + (P_f - P_s)Q_2)/(3.6 \times 10^6 \cdot Q_1 \eta_e \eta_p) \quad (9)$$

$$P^{DD} = \bar{P} + 0.05 \cdot P^{DDN} \quad (10)$$

$$Q_f = Q_p + Q_c = Q_1 + Q_2 \quad (11)$$

$$\Delta P_{fc} = P_f - P_c \quad (12)$$

where the energy balances are constrained to (9)–(10), where $\bar{P}$ denotes the desalination motor power, η_e and η_p represent the motor and pump efficiency, P^{DDN} and P^{DD} represent the rated and actual desalination power. The material balance is given in (11). The pressure difference constraint between the head and end is written in (12).

According to (9)–(12) and membrane parameters in Table 2, we can obtain the following equation:

$$P^{DD} = 0.0030507 \times [P_f/Y - 0.855(P_f - \Delta P_{fc})(1/Y - 1)]/(\eta_e \eta_p) \times [P_f - 0.035(32.35/Y - 16.175)^{1.7} - 0.0005] + 328.2(\ln(1 - Y)/Y \times \exp(0.7(1 - (1 - Y)^{1/7}))) + 0.05P^{SDEN} \quad (13)$$

On this basis, the DD devotes to maximizing the desalination output by optimizing the P_f and P_f , as follows:

$$\max Q^{DD} \quad (14)$$

$$s. t. \quad P_f \leq 1001.5 \text{ psi} \quad (15)$$

$$Y \leq 45\% \quad (16)$$

(8) and (13)

where (14)–(16) express the mechanical strength constraints.

Based on the MATLAB platform, the desalination characteristic can be solved using Fmincon solver. The power-output relationship is shown in Fig. 3.

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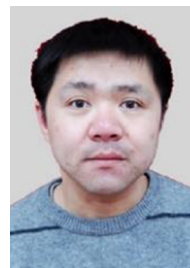
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Quan Sui received the B.Sc. degree in electrical engineering from Huazhong University of Science and Technology (HUST) in 2017. He is studying for a Ph.D at HUST. His research interests are microgrid planning and scheduling as well as optimal power system.



Fanrong Wei received the Ph.D. degree in electrical engineering, Huazhong University of Science and Technology (HUST). His researches mainly focus on optimal power system/microgrid scheduling and protective relay.



Xiangning Lin is currently a Professor at Huazhong University of Science and Technology (HUST). He received the M.Sc. and Ph.D. degrees in electrical engineering from HUST, Wuhan, China. His research interests are modern signal processing and power system protective relaying.



Zhengtian Li received the B.Sc. degree from Wuhan University in 2002 and the Ph.D. degree from the Huazhong University of Science and Technology (HUST) in 2011. Currently, he is an Associate Professor at HUST. His research interests are digital protection relaying.